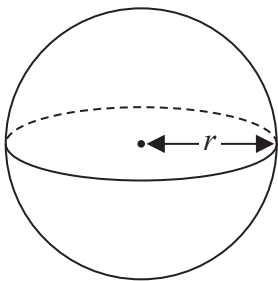
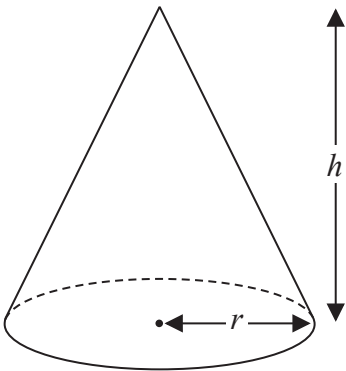


24 Here is a solid sphere and a solid cone.



Volume of sphere = $\frac{4}{3} \pi r^3$



Volume of cone = $\frac{1}{3} \pi r^2 h$

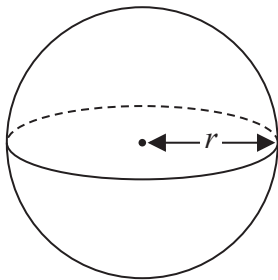
All measurements are in cm.

The volume of the sphere is equal to the volume of the cone.

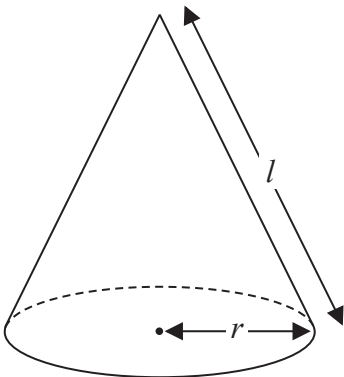
- (a) Find $r:h$
Give your answer in its simplest form.

(2)

Here is a different solid sphere and a different solid cone.



Surface area of sphere = $4\pi r^2$



Curved area of cone = $\pi r l$

All measurements are in cm.

The surface area of the sphere is equal to the **total** surface area of the cone.

- (b) Find $r:h$
Give your answer in the form $1:\sqrt{n}$ where n is an integer.

(4)